Chapter 1

1 Classify each of the beams/structures shown below as statically determinate or indeterminate. If statically indeterminate, report the number of degrees of indeterminancy. The beams/structures are assumed to be subjected to known external loadings acting at known positions. **4**

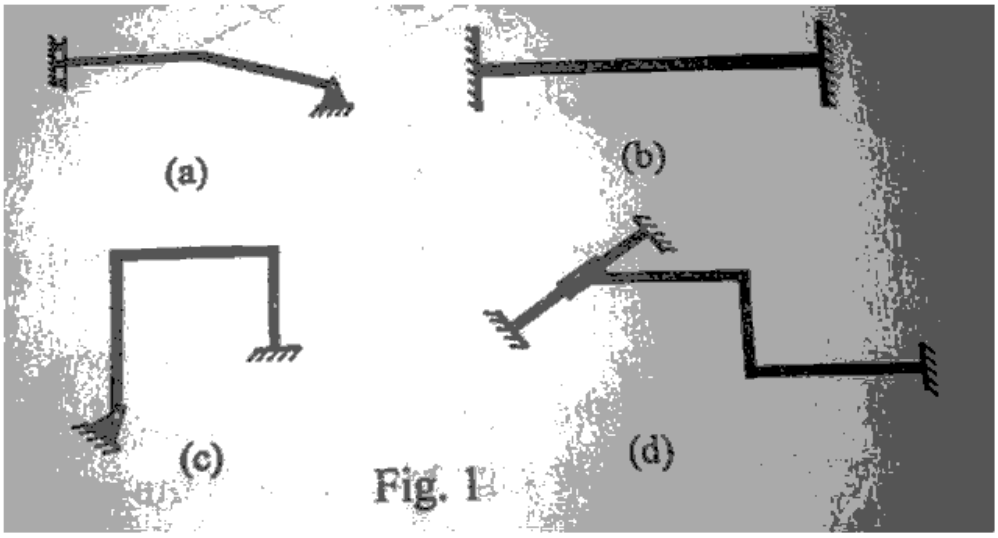
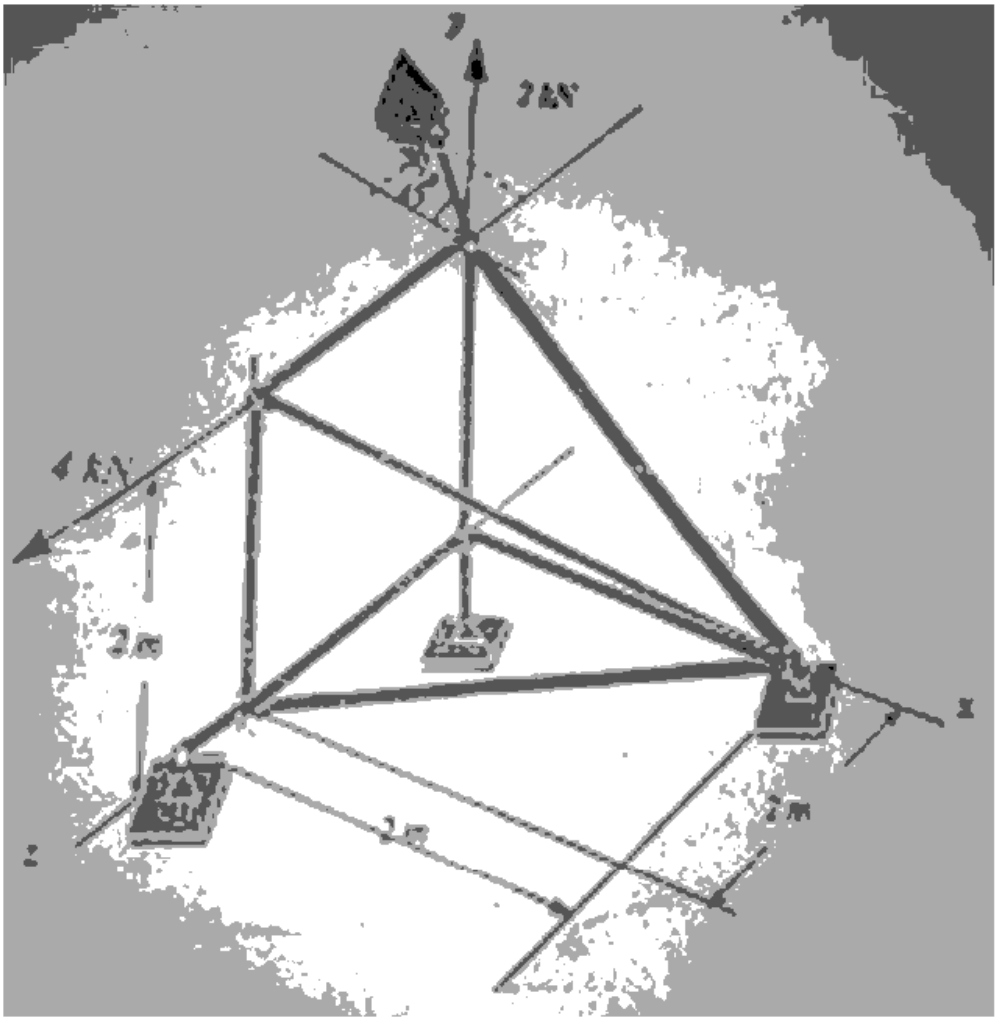
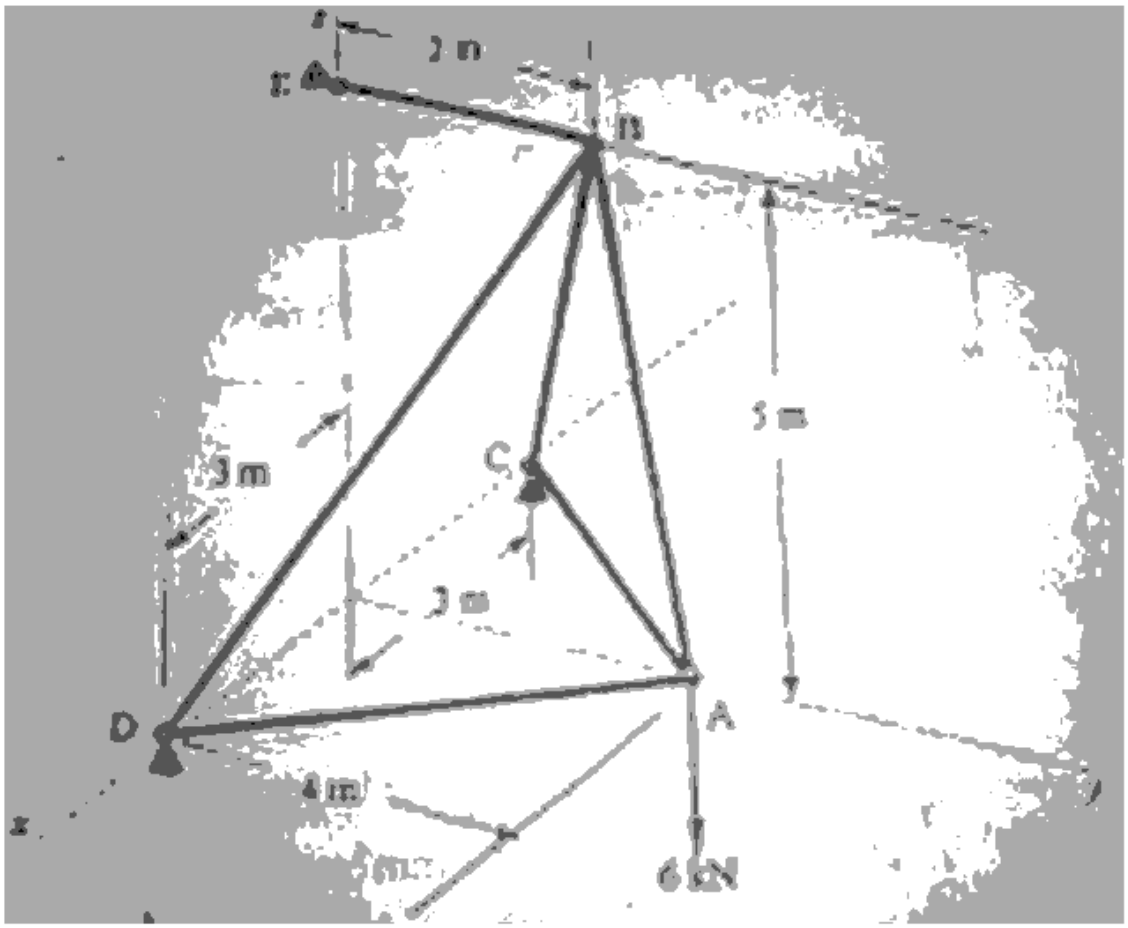


Fig. 1

2 Determine the force in each member of the space truss shown in fig.2. Indicate if the members are in tension or compression. **12**



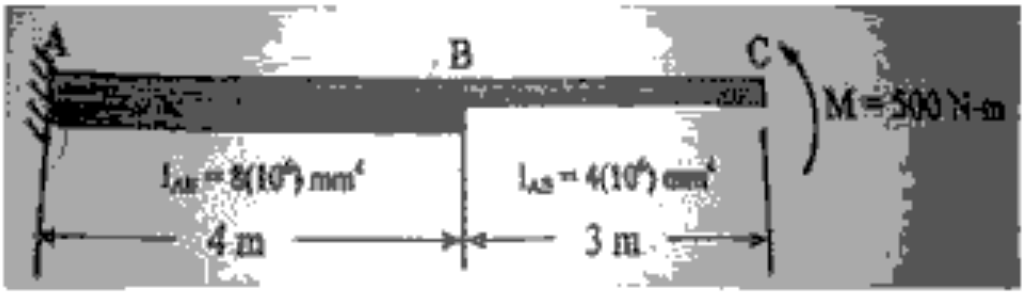
3 Determine the forces in each member of the space truss shown below and state if the members are in tension or compression. The support reaction at E acts along member EB. **12**



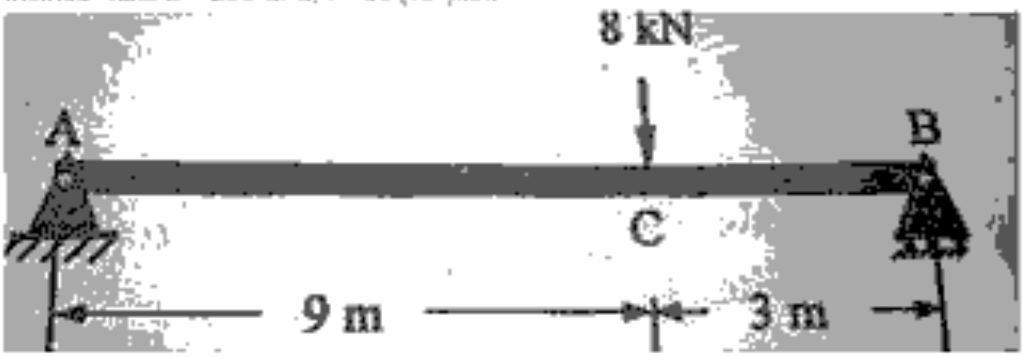
Chapter 2

1 Draw conjugate beams for the real beams shown in fig **4**

2 Determine the slope and deflection at point B and C of the beam shown in fig. Values for the moment of inertia of each segment are indicated in the figure. Take E = 200 GPa. State the us the moment area theorem I and II **12**



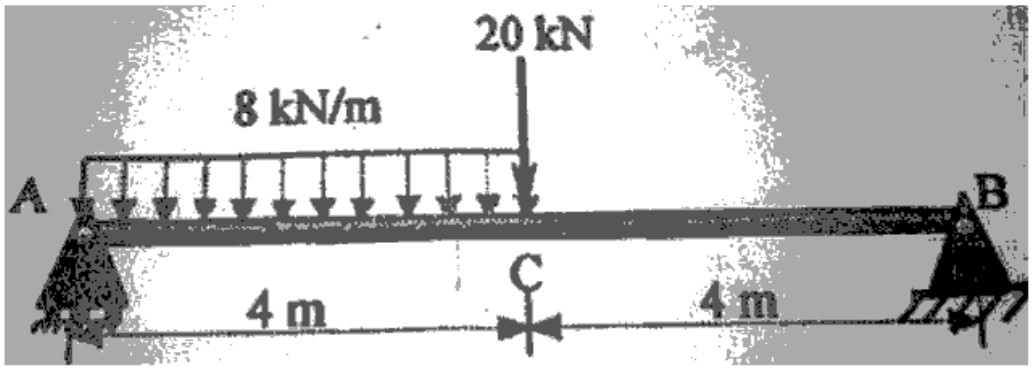
3 Determine the maximum deflection and deflection at C of the steel beam shown in fig using conjugate beam method. State the theorem associated with conjugate beam method. Take E = 200 GPa, I = 60 (10^6)mm^4 **12**



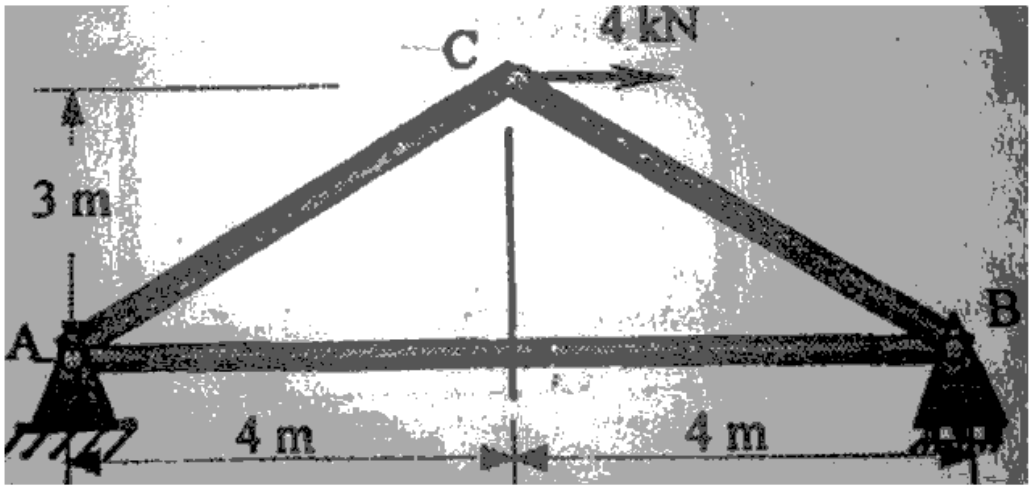
Chapter 3

1 Write strain energy relations due to axial load, bending, shear and torsion. **4**

2 Determine the displacement of point C of the steel beam shown in fig. State and use Castigalino's theorem. Take **12**
E = 200 GPa, I = 150 (10^6)mm^4.



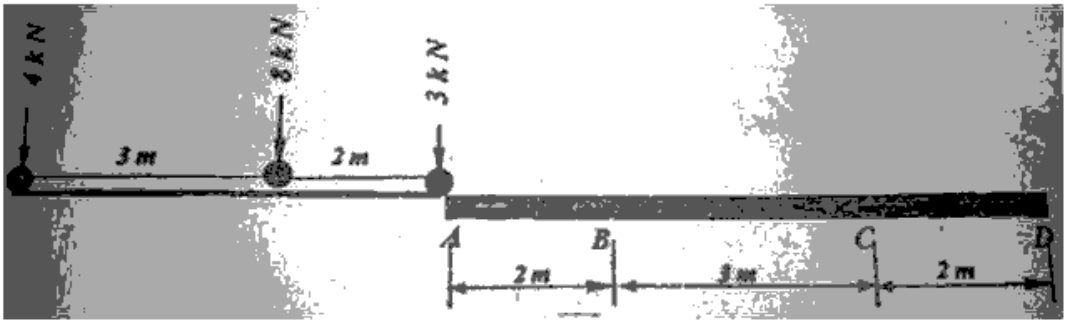
3 Determine the vertical displacement of joint C of the truss shown in fig, using Castiglino's theorem. Take E = 200GPa, A = 400 mm^2. **12**



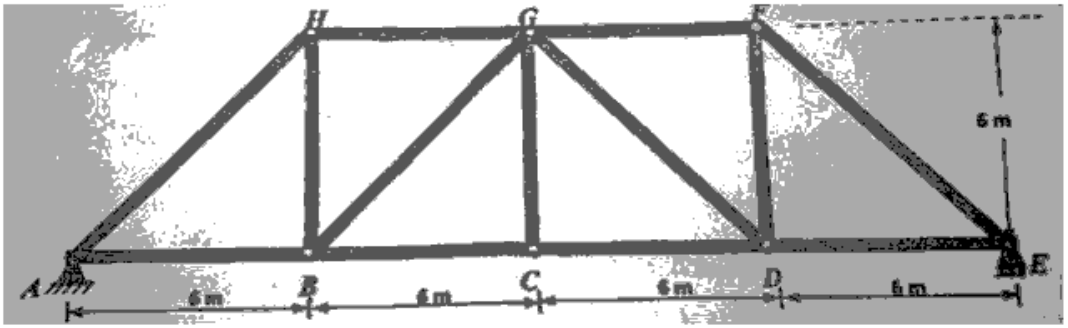
Chapter 4

1 Clearly state the difference between construction of an influence line and construction of a shear and moment diagram. Also state the Muller-Breslau principle. **4**

2 Determine the maximum positive moment created at point B in the beam simply supported at points A and C shown in fig, due to the wheel loads of the crane. Also, determine the absolute maximum moment. **12**



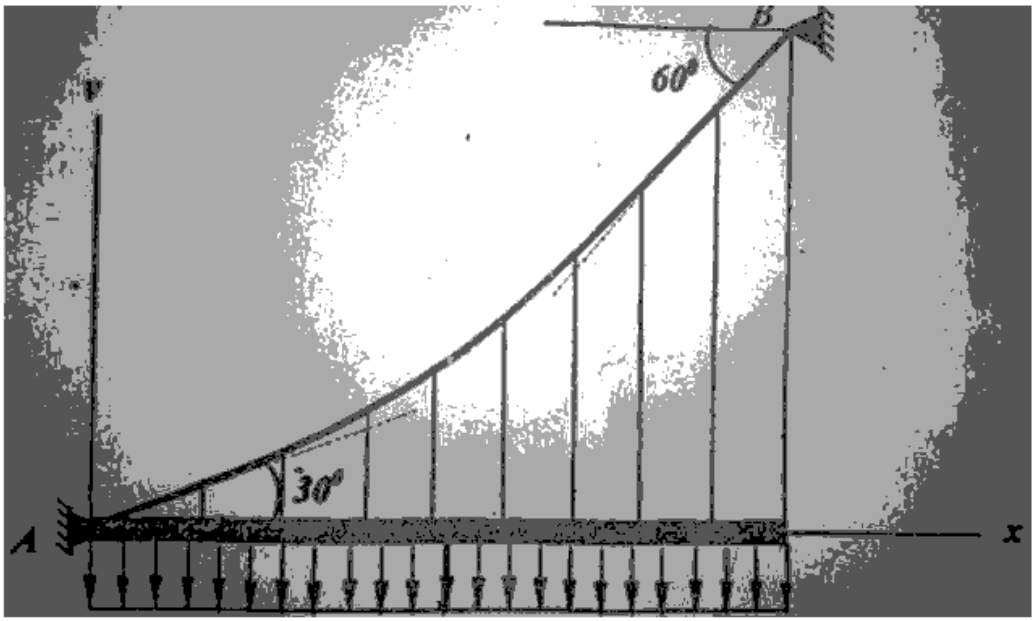
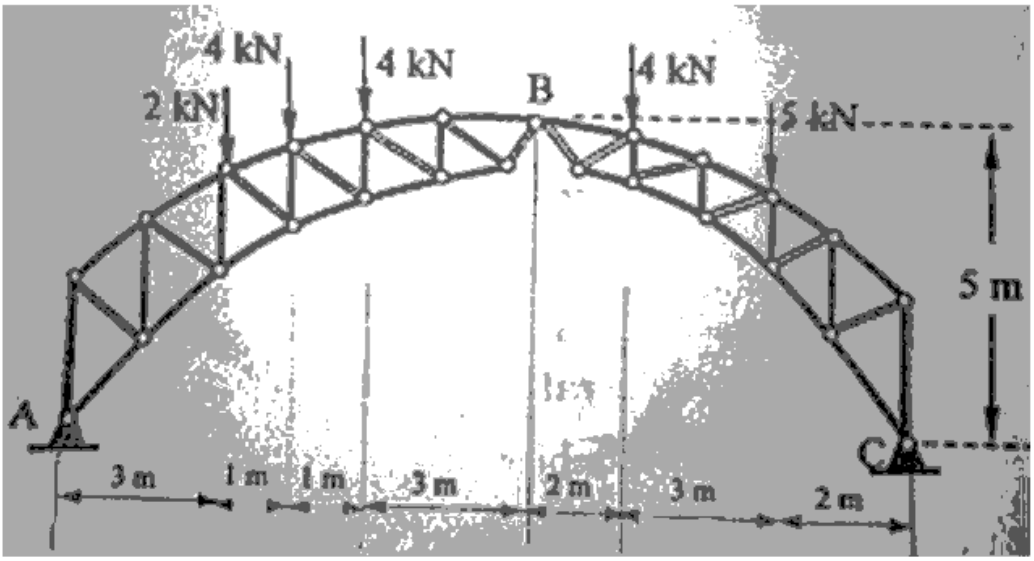
3 Draw influence line for the force in member GB and CG of the bridge truss shown in fig. **12**



Chapter 5

1 Explain how the cables and arches are different from other structural elements. **4**

2 The cable AB is subjected to a uniform loading of 200 N/m. If the weight of the cable is neglected and the slope angles at point A and B are 30^0 and 60^0 respectively, determine the curve that defines the cable shape and the maximum tension development in the cable. **12**



3 Determine the resultant forces at the pins A, B and C of three hinged arched roof truss shown in fig. **12**